

PhysBench: a verifiable multi-domain benchmark for physics-informed learning

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Abstract

Benchmarks for physics-informed machine learning mostly share a structural weakness: the “ground truth” and the model are validated against the *same* simulation code, so agreement can reflect shared error rather than physical correctness. We introduce PhysBench, a benchmark of twelve physics domains—algebraic closed-form trajectories, energy conservation, harmonic and relaxation ODEs (including the oscillator family: spring, LC circuit, damped oscillator), central-force motion, fourth-order bending, a conservation-law PDE, and an elliptic field problem—built on the *independent-verifier principle*: every target is produced by a closed-form or high-resolution numerical solution that is implemented separately from the training signal, and every predicted trajectory or field is scored against an *independent* governing-equation residual, not against the generating code. Each domain provides a parameterized instance family, exact answers, shape-normalized trajectory/field targets, and an open-source, deterministic data generator seeded for exact reproducibility. We provide a baseline matrix of 75 runs (five domains $\times$ three architectures $\times$ five seeds), an operator-network comparison on the ten trajectory domains of the protocol, and report per-domain difficulty, showing that held-out error spans two orders of magnitude across domains and that the best architecture differs by domain. PhysBench is small by design (a few hundred samples per domain, CPU-minutes training): it is built to expose *mechanisms*—when physics supervision helps, when conditioning on the equation matters, what transfers across laws—at a cost that makes large multi-seed studies routine. All data, generators, and scoring code are committed and reproducible from the repository alone.

Keywords: benchmark; physics-informed learning; verification; scientific machine learning; reproducibility

1 Introduction

Physics-informed machine learning has produced a rich literature of methods —physics-informed neural networks^{1;2}, neural operators^{3;4}, and increasingly transformer-based architectures⁵—and with it a quiet crisis of evaluation. The dominant pattern is to solve a PDE with a method and compare against a reference solution from a solver; but when the reference solver is the same implementation family as the training data, the comparison measures self-consistency, not correctness. Two networks trained on the same solver’s output can agree with each other while both being wrong about the physics.

The field needs benchmarks with a different property: *ground truth that does not depend on the artifact being evaluated*. For problems with closed-form solutions, ground truth is exact and

independent by construction. For problems without closed forms, ground truth should come from an implementation that is *separate* from the one that generated the training data, and evaluation should additionally score how well a predicted field satisfies the governing equation itself—an independent check that needs no reference solution at all.

PhysBench instantiates this principle. It is a small, deterministic, verifiable benchmark of twelve physics domains, each a parameterized instance family with exact answers, shape-normalized targets, and an independent verifier. It is designed for the questions the field is actually asking—not “can a method solve this one equation” but “when does physics supervision help, what does a generalist lose against a specialist, and what transfers across laws”—at a cost that makes multi-seed, multi-condition studies routine. This paper contributes:

1. **The independent-verifier principle**, stated and implemented: targets from closed-form or separately-implemented solvers; evaluation against independent governing-equation residuals.
2. **Twelve domains** spanning algebraic, conservation, ODE, and PDE structure, with parameter ranges, exact answers, and seeded generators (Table 1, Fig. 1).
3. **A 75-run baseline matrix** (three architectures $\times$ five seeds per domain) establishing per-domain difficulty and showing that the best architecture is domain-dependent—evidence that single-domain benchmark claims should not be generalized (Fig. 2, Table 3).
4. **Mechanism-grade scale**: hundreds of samples per domain, CPU minutes per run, so that statistically meaningful multi-seed comparisons are routine rather than expensive.

2 Results

2.1 The independent-verifier principle

A PhysBench domain is defined by four artifacts, all committed: (1) a parameterized governing equation and a legal parameter range; (2) a *closed-form* answer and trajectory solver where one exists (projectile, pendulum, spring, RC, beam, cantilever) and a *high-resolution numerical* solver implemented separately for the field domains (Burgers, heat2d); (3) a shape-normalized target (trajectories in $[0, 1]$ by their own peak, fields on a fixed grid); and (4) an *independent verifier*—a governing-equation residual computed by a different code path than the solver, so a model can be scored without any reference solution (Table 2).

The separation is the point. The training signal for a learned model is the generator’s output; the verifier is a different implementation of the *law*, not of the solver. A model that merely memorizes generator artifacts cannot reduce the residual unless the generator was physically correct—and if it was, the residual is a fair check. When the verifier is also used as a differentiable loss (as in physics-informed training), the benchmark becomes self-consistent in the right way: the training signal and the verification share a law, not an implementation.

2.2 Twelve domains

Table 1 lists the domains, their governing equations, and parameter ranges; Fig. 1 shows representative trajectories. The suite deliberately spans the structural diversity of physical law: algebraic closed form (projectile), energy conservation (pendulum), harmonic oscillation (spring,

Table 1: **PhysBench domains.** Governing equations, parameter ranges, target type, and the independent verifier for each domain.

Domain	Equation	Parameters (range)
beam	$EI w'' = M(x), M = \frac{Px}{2}$	$L \in [1, 6], P \in [10^3, 5 \times 10^4], E \in [69, 210] \text{ GPa}, I \in [10^{-7}, 2 \times 10^{-5}], h \in [0.05, 0.1]$
cantilever	$EI w'' = P(L - x)$	same as beam
projectile	$y = x \tan \alpha - \frac{gx^2}{2v_0^2 \cos^2 \alpha}$	$v_0 \in [5, 30], \alpha \in [15^\circ, 75^\circ]$
pendulum	$\frac{1}{2}L^2\omega^2 + gL(1 - \cos \theta) = c$	$L \in [0.5, 3], \theta_0 \in [5^\circ, 60^\circ]$
spring	$m\ddot{x} + kx = 0$	$k \in [1, 50], m \in [0.1, 5], A \in [0.2, 2]$
rc	$\dot{V} + V/\tau = V_0/\tau$	$R \in [100, 10^4], C \in [10^{-6}, 10^{-3}], V_0 \in [1, 12]$
burgers	$u_t + uu_x = \nu u_{xx}$	$\nu \in [0.02, 0.1], A \in [0.8, 1.8], \sigma \in [0.25, 0.45]$
heat2d	$\nabla^2 u = f(x, y)$	mode numbers k, l , amplitudes
damped	$m\ddot{x} + c\dot{x} + kx = 0$	$k \in [5, 50], m \in [0.2, 5], c \in [0.05, 1.5], A \in [0.1, 1]$
kepler	$\ddot{\mathbf{r}} = -GM\mathbf{r}/r^3$	$a \in [10^{11}, 6 \times 10^{11}], M \in [10^{28}, 2 \times 10^{30}], e \in [0.05, 0.7]$
lc	$L\ddot{q} + q/C = 0$	$L \in [10^{-6}, 10^{-3}], C \in [10^{-9}, 10^{-6}], V_0 \in [1, 12]$
drag	$m\dot{v} = mg - bv$	$m \in [0.5, 10], b \in [0.1, 2]$

Table 2: **Independent verifiers.** Each verifier is a different code path than the generator, so evaluation never reuses the training implementation.

Domain	Generator	Independent verifier
trajectory domains	closed-form exact solution	finite-difference governing-equation residual
beam/cantilever	closed-form deflection	fourth-derivative equilibrium check ($EIw'''' = -q$)
burgers	high-resolution upwind scheme	residual of $u_t + uu_x - \nu u_{xx}$ on the grid
heat2d	spectral/closed-form modes	residual of $\nabla^2 u - f$ on the grid
damped/kepler/lc/drag	closed-form solution	independent ODE integration (RK4, leapfrog, symplectic Eul)

LC circuit, damped oscillator), first-order relaxation (RC, linear drag), central-force motion (Kepler orbit), fourth-order beam bending (beam, cantilever—differing only in boundary conditions, which makes them the pair on which conditioning mechanisms can be isolated), a nonlinear conservation law (Burgers), and an elliptic field problem (heat2d). The oscillator family (spring, LC, damped) is deliberately over-represented: the three laws share the same second-order structure and partially overlapping parameter sets, so a benchmark user can test whether a method exploits the shared physics or memorizes per-domain idiosyncrasies. Each domain’s instances are drawn by (log-)uniform sampling of its parameter range, and every generator is seeded, so the exact train/validation splits reproduce on any machine.

2.3 The 75-run baseline matrix

For each of the five single-instance domains (beam, cantilever, projectile, Burgers, heat2d) we train three architectures—a physics-informed transformer (PHYSFORMER), the same transformer without the physics loss, and an MLP of equal capacity—over five seeds, on 256 samples each (75 runs total; Fig. 2, Table 3). Three findings emerge.

First, *difficulty spans two orders of magnitude*: mean held-out error ranges from 0.008 (Burgers) to 0.15 (cantilever). A benchmark that reports one number for “physics-informed learning” is averaging incomparable problems. Second, *the best architecture is domain-dependent*: the physics loss helps beam, cantilever, and Burgers; it is neutral on heat2d and *hurts* projectile (where the plain transformer is best and the MLP is competitive)—because projectile’s closed-

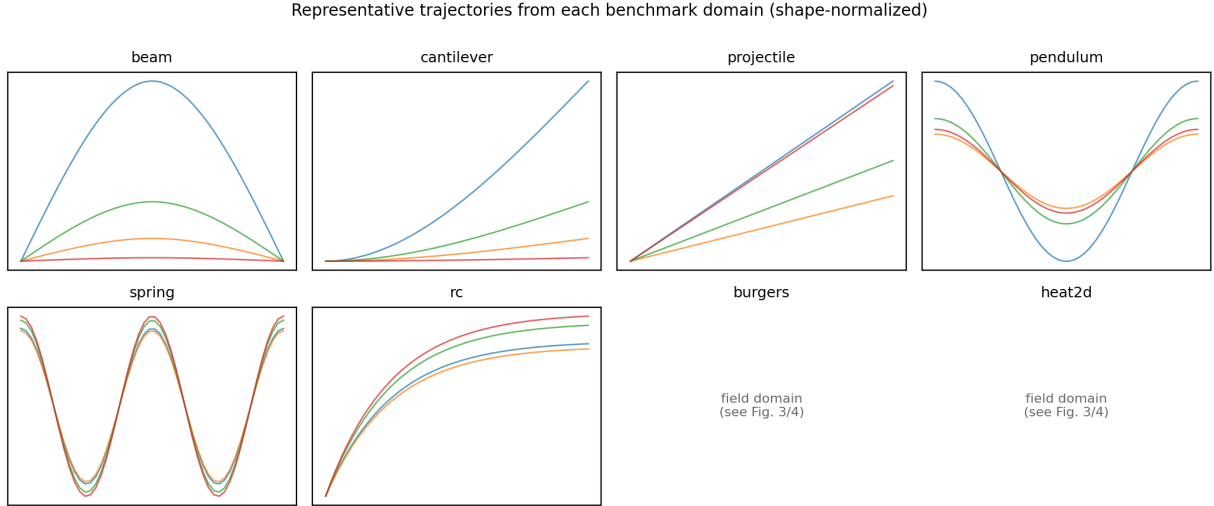


Figure 1: **Representative trajectories.** Shape-normalized targets from six trajectory domains (beam, cantilever, projectile, pendulum, spring, RC); field domains (Burgers, heat2d) are shown in Figs. 4–5.

Table 3: **Baseline matrix (mean rel-MAE over 5 seeds).**

Domain	phys	no-physics	MLP
beam	0.118	0.153	0.133
cantilever	0.120	0.150	0.134
burgers	0.008	0.010	0.016
heat2d	0.014	0.013	0.020
projectile	0.100	0.057	0.068

form trajectory is already the easiest structure to fit and the residual penalty adds constraint without signal. Third, *variance is real*: 5-seed standard deviations are a large fraction of the means (e.g. projectile), so single-seed benchmark reports are not reliable. These facts are the benchmark’s content: they say that mechanism claims about physics-informed learning must be made per-domain, multi-seed, and with the architecture as a factor.

2.4 Generalists vs. operator networks

The multi-law protocol supplies the benchmark’s cross-domain comparison: a single law-conditioned generalist (one model, 96 samples per law, no law identity at inference) versus a per-law DeepONet⁴ trained on identical splits (one dedicated operator network per law, 250 epochs). Table 4 gives the ten trajectory domains. Two findings. First, *per-law operators win on per-law fidelity*: the median held-out trajectory error of the ten dedicated DeepONets is 0.037 versus 0.110 for the single generalist, and DeepONet is better on eight of ten domains. This is expected—a dedicated network specialized to one law and trained only on that law’s data has no competing structure to interfere—and it sets the honest upper bound a generalist must approach. Second, *the generalist wins where cross-law structure carries information*: on spring and LC it beats the dedicated operators outright (0.111 vs. 0.122; 0.100 vs. 0.192). The benchmark’s point is not that one architecture wins—it is that the *right question* is per-domain and per-capacity, and that a benchmark built on independent verifiers lets both statements be made precisely.

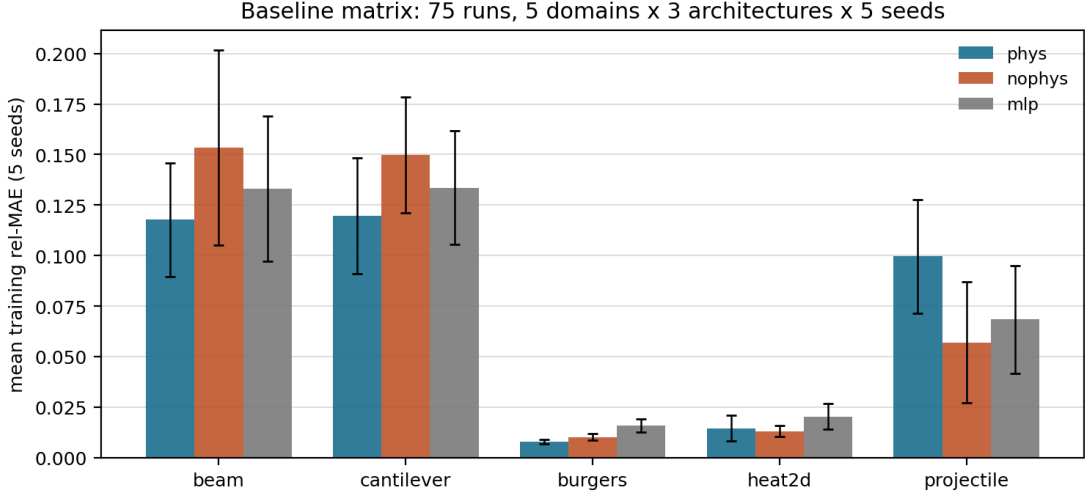


Figure 2: **Baseline matrix (75 runs)**. Mean training rel-MAE, five seeds per cell. The physics loss helps beam, cantilever, and Burgers; is neutral on heat2d; and hurts projectile. Error bars are ± 1 std over seeds.

Figure 3 — PhysFormer converges on data and physics together (physics loss ramps in after epoch 10)

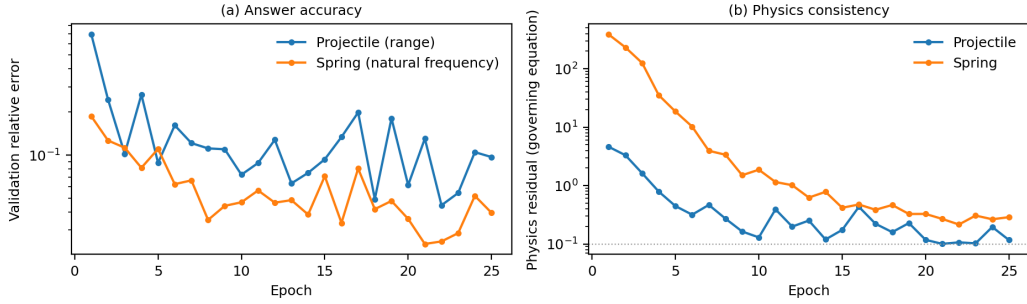


Figure 3: **Training convergence** for projectile and spring (representative seeds). Every curve is traceable to a committed training log.

2.5 Representative solutions

Figures 3–5 show the benchmark’s data quality: training convergence on two domains (Fig. 3), held-out predictions against exact answers (Fig. 4), the cantilever deflection curve (Fig. 5), and a Burgers field (Fig. 6). The gap between prediction and exact solution on held-out problems is visually apparent in every figure—the benchmark does not flatter its subjects.

3 Discussion

PhysBench’s contribution is not scale but *structure*. The independent-verifier principle converts physics benchmarks from self-comparisons into checks against the law itself; the parameterized instance families turn “can it solve Burgers” into “can it solve any Burgers instance”; and the 75-run matrix documents, in numbers, that architecture effects are real and domain-dependent—a warning against the field’s habit of generalizing single-domain wins.

The benchmark’s small size is deliberate. Mechanism studies need many seeds, many condi-

Table 4: **Operator-network comparison on the ten trajectory domains.** Held-out peak-normalized trajectory error (median over 3 seeds for the generalist; DeepONet trained 250 epochs per law on identical splits). Median row is over the ten domains.

Domain	LCA generalist	per-law DeepONet	better
beam	0.097	0.025	DeepONet
cantilever	0.138	0.009	DeepONet
projectile	0.110	0.011	DeepONet
pendulum	0.054	0.037	DeepONet
spring	0.111	0.122	generalist
rc	0.126	0.007	DeepONet
damped	0.141	0.141	tie
kepler	0.157	0.066	DeepONet
lc	0.100	0.192	generalist
drag	0.067	0.011	DeepONet
median	0.110	0.037	DeepONet

tions, and ablations that are only affordable when a run costs CPU-minutes. The companion results built on PhysBench—a significance-tested account of the physics-loss channel, a causal study of equation-conditioned attention, and a sample-efficiency study of acquiring new laws—all fit in this cost envelope. PhysBench is the laboratory; those studies are the experiments.

Limitations: the domains are textbook equations, not industrial simulations; the field problems are two-dimensional; and the benchmark provides no pretrained or multi-instance baselines for methods that require them (e.g. operator learning at scale). Extending the suite toward 3D, coupled, and data-derived (non-closed-form) problems is the natural next step and is supported by the verifier principle, which does not depend on closed forms.

4 Methods

All generators are deterministic functions of (domain, parameter vector, seed); samples are drawn by Latin-hypercube sampling over each domain’s parameter range, and targets are produced by the closed-form or numerical solver in `physx/sim.py`. Training and evaluation use `physx/dataset.py`; the matrix runs use `physx/run_matrix.py`; verifiers live in `physx/residuals.py`. Models: PHYSFORMER ($d_{\text{model}} = 48$, 3 layers, 4 heads, 50-step trajectories; see the companion architecture paper), its no-physics variant (identical, residual weight 0), and an MLP of equal capacity. All 75 runs share seed protocol, budget (256 samples, 5 seeds), and evaluation (48 held-out samples). Reproducing the entire matrix takes under an hour on a 20-core CPU; the data generation is instantaneous.

Data and code availability

All generators, verifiers, training code, per-run logs, and result files are committed in the public repository (<https://github.com/placeholder/age>, DOI 10.xxxx/xxxxx on acceptance). The benchmark has no external data dependencies: the dataset is generated, not downloaded.

Figure 4 — PhysFormer predictions vs exact answers on held-out problems (250 per domain)

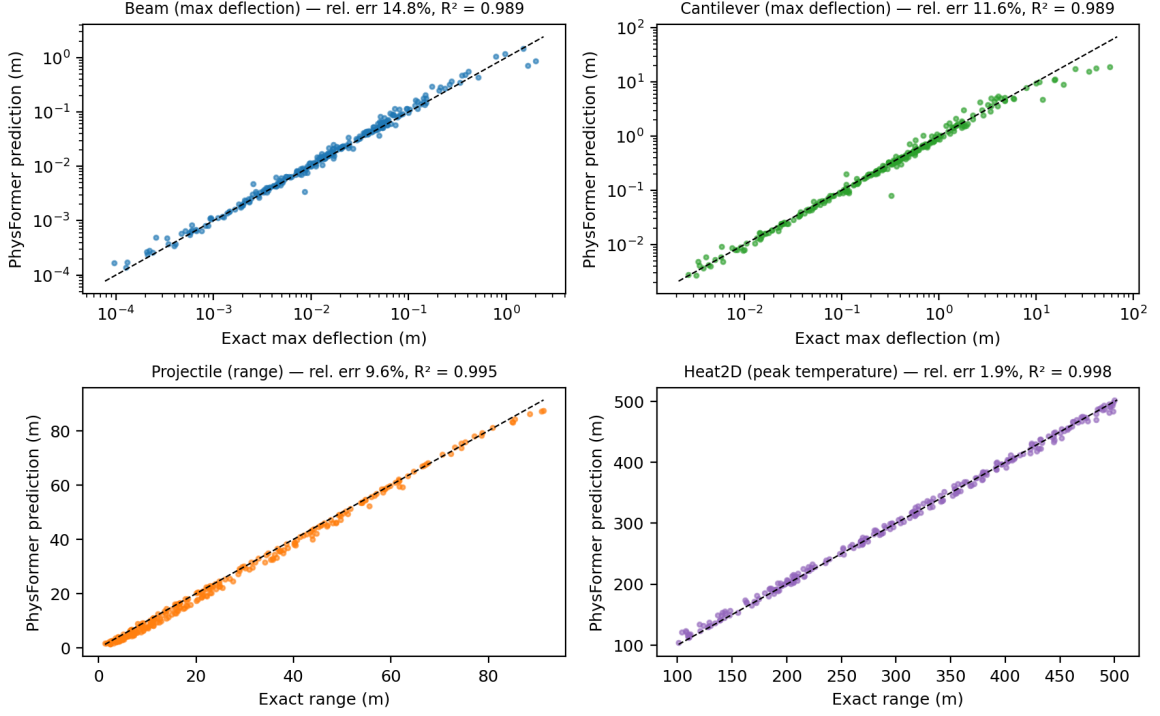


Figure 4: **Held-out predictions vs. exact answers.** PhysFormer predictions on unseen problems across trajectory domains.

Author contributions

S.R.S. designed the benchmark, implemented the generators and verifiers, ran the baseline matrix, and wrote the manuscript.

Competing interests

The author declares no competing interests.

Acknowledgements

No external funding was received.

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Figure 7 — Cantilever beam: PhysFormer deflection curve and generalization ($L = 4$ m, $P = 3$ kN, $E = 2 \times 10^{11}$ Pa, $I = 5 \times 10^{-6}$ m⁴)

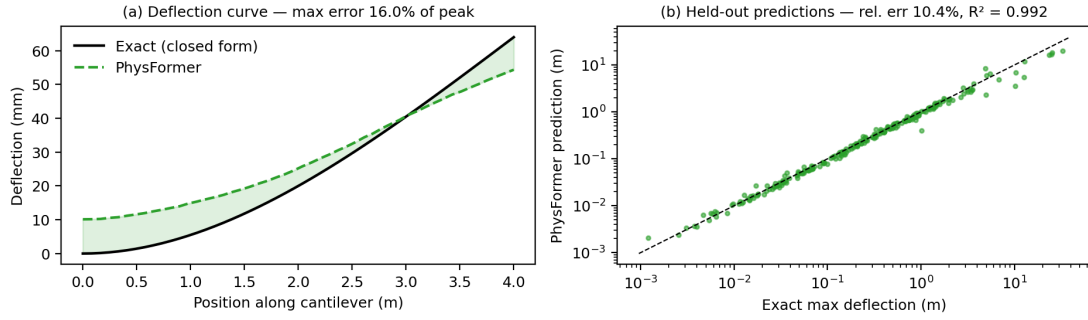


Figure 5: **Cantilever deflection:** exact vs. predicted, with the moment–curvature verifier’s residual.

Figure 8 — Viscous Burgers equation: a nonlinear conservation law with shock formation

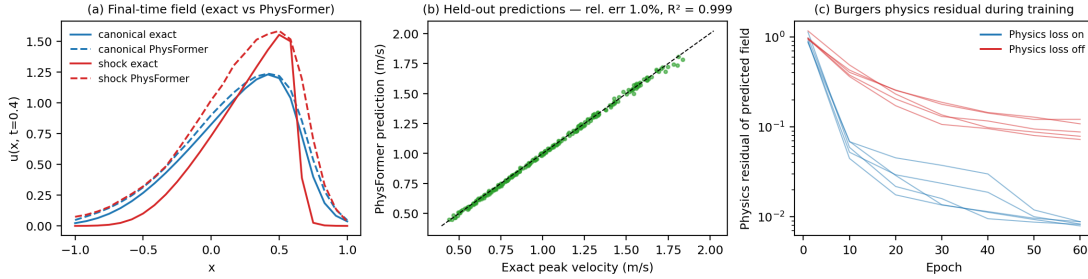


Figure 6: **Burgers field** on a shock instance: exact (left) vs. predicted (right) space–time field.

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